

Privacy-Friendly Adaptation of Vision Transformers for Communication and Latency-Efficient Private Inference

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Abstract

The advent of machine learning as a service (MLaaS) has necessitated secure multi-party computation (MPC)-based private inference (PI) to address the privacy concerns that arise when web servers offer query inference services to users. However, the formal privacy protection incurs substantial communication and latency overheads, particularly for large models such as vision transformers (ViTs). Existing methods either ignore the inherent attention dependencies or naively extend CNN optimizations to ViTs despite their structural discrepancies, resulting in sub-optimal performance. In this paper, we co-design the MPC and architectural properties of ViT and propose SecViT, an efficient and secure inference framework that automatically adapts ViTs into privacy-friendly counterparts with optimal attention configurations at different layers and tokens under MPC, balancing model capability and efficiency. SecViT features an MPC-efficient, layer-dependent and load-balancing attention representation adapter to facilitate feature reuse across multiple highly correlated layers without impacting accuracy. To further reduce the inference cost, SecViT also develops fine-grained composite attention and activation approximation algorithms to achieve superior accuracy-efficiency trade-offs. Experiments show that SecViT reduces communication by 6.0× and latency by 4.6× with iso-accuracy over MPCViT, and improves accuracy by 4.92%

with 1.6× lower latency over PriViT on Tiny-ImageNet. Compared with state-of-the-art PI protocols, SecViT further achieves 10.0× communication saving and 7.9× latency reduction over BumbleBee.

CCS Concepts

• Security and privacy → Privacy-preserving protocols; • Computing methodologies → Neural networks.

Keywords

Privacy-Preserving Machine Learning, Secure Multi-Party Computation, Vision Transformer, Secure and Efficient Algorithms

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1 Introduction

The growing adoption of machine learning (ML) has reshaped a wide range of web applications such as face authentication [42], online recommendation [44], healthcare assistance [55], etc. However, the intensive data and computational resources required for training powerful ML models hinder their practical deployment for personal use. The users have no choice but to send their personal data to web service providers for remote inference, known as the market of machine learning as a service (MLaaS). Privacy remains a critical concern in the implementation of MLaaS since both the client and the server are reluctant to fully disclose their sensitive data or model parameters to each other. To address this dilemma, secure multi-party computation (MPC)-based private inference (PI)

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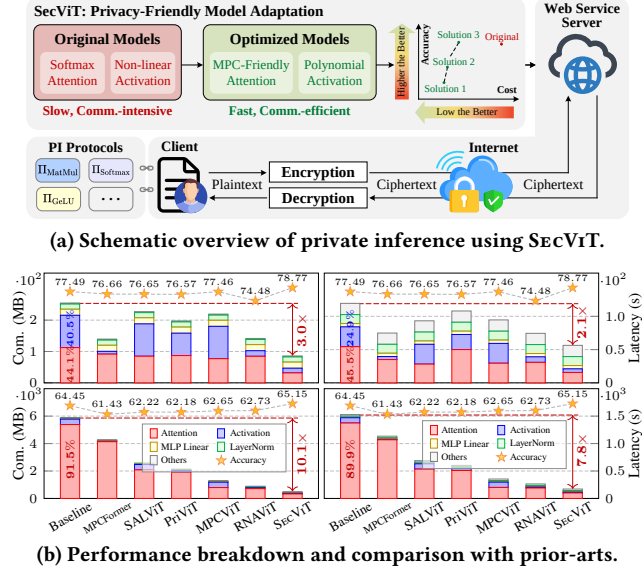


Figure 1: (a) Illustration of SECViT framework that adapts ViTs for fast and communication-efficient private inference. (b) Performance comparison under a WAN setting. Top (resp. Bottom): The baseline model is a 7 (resp. 9)-depth ViT with 4 (resp. 12) heads on CIFAR-100 (resp. Tiny-ImageNet). All results are evaluated using the same baseline and MPC library for a fair comparison. SECViT significantly outperforms the prior works in terms of both efficiency and accuracy.

allows the client and the server to conduct cryptographically secure inference jointly on the encrypted input and model, without leaking any knowledge of the involved data (Figure 1a).

The formal privacy protection of PI comes at the price of significantly higher communication and latency costs compared to its plaintext counterpart, and struggles to scale efficiently to large models such as vision transformers (ViTs). To bridge this efficiency gap, tremendous efforts have been dedicated to designing customized cryptographic protocols [14, 30] and MPC-optimized network architectures [52] for convolutional neural networks (CNNs). However, directly applying these CNN optimizations to ViTs often leads to sub-optimal performance because of their distinct structural and MPC properties, particularly the multi-head attention (MHA) and GeLU functions which are absent in CNNs but essential in ViTs [8]. We profile the communication and latency breakdown of ViTs in Figure 1b, which reveals that attention operations dominate up to 91.5% of communication and 89.9% of latency overheads on Tiny-ImageNet due to the expensive reciprocal and exponential within the Softmax, while GeLU also incurs non-negligible inference cost on CIFAR-100, both of which hinder the MPC efficiency of ViTs.

Apart from the Softmax attention, several linear attentions exist [27, 37, 40, 46]. Unfortunately, these approaches remain infeasible for PI because they heavily rely on normalization or matrix multiplication to approximate Softmax which are expensive in MPC. To alleviate this, recent efforts [3, 6, 22, 51, 54] have explored PI-efficient ViTs by simplifying the costly MHA and GeLU. For attention optimization, most of them either reduce the input dimension of Softmax or replace Softmax with more efficient variants. For

GeLU optimization, these approaches selectively linearize GeLUs into identity functions, thus reducing the inference cost.

Despite their promises, current methods still struggle with the prohibitively high inference cost. The most recent method PriViT requires a communication of 4.2GB and a latency of 1116s for each inference at the scale of Tiny-ImageNet, or 1.0GB and 335s with 4.22% accuracy loss. We argue that these methods yield limited efficiency because of the following constraints (i) **the ignorance of attention correlation** remains the efficiency optimization of Softmax approximation not yet fully explored; (ii) **the lack of a performant yet MPC-efficient attention approximation** that trades off better efficiency and accuracy; (iii) **GeLU exhibits different numerical properties compared to ReLU**, simply linearizing GeLU as done in CNNs may impair the model capability of ViT.

In response, we propose SECViT, an efficient and secure inference framework that tailors performant and MPC-efficient ViTs for the needs of PI. SECViT features an MPC-efficient attention representation adapter to facilitate feature reuse across multiple MHA blocks (**addressing (i)**), and develops an efficient yet performant Scale ReLU attention to approximate Softmax attention (**addressing (ii)**), achieving better accuracy-efficiency trade-offs. SECViT also employs a fine-grained polynomial replacement guided by feature-level distillation to boost the activation efficiency without impacting the inference accuracy (**addressing (iii)**).

In summary, our main contributions include:

- To our best knowledge, SECViT is the first attempt that successfully leverages attention correlation for efficient PI, and develops a suite of MPC-friendly approximations, yielding a new family of secure and efficient ViTs.
- We reformulate the attention structure for MPC, steered by a two-stage search algorithm that unifies efficient attention reuse and composite attention approximation to improve both latency and communication efficiency.
- To further reduce the inference cost, we present a sensitivity-aware differentiable activation replacement algorithm coupled with a learnable quadratic approximant to achieve better trade-offs between accuracy and efficiency.
- SECViT shows impressive performance by 1.6 ~ 11.4× communication and 1.4 ~ 7.9× latency reductions with higher or iso-accuracy over prior network-level methods. SECViT is also compatible with state-of-the-art (SOTA) PI protocols and can be applied on top of them for further efficiency gains.

2 Related Works

2.1 Private Inference

The private inference is pioneered by CryptoNets [9] on CNNs with the cryptographic primitive of either homomorphic encryption (HE) [1, 16, 17], MPC [7, 18, 38] or both [48, 50, 53] throughout the evaluation of linear and non-linear operations. Previous approaches [5, 10, 15, 21, 23, 32–34] have observed that ReLU dominates the PI overhead and aim to develop ReLU-efficient CNNs. However, for ViTs, the main bottleneck becomes the attention and GeLU components, which are absent in CNNs and thus under-explored. Constrained by the limited efficiency gains achievable solely through protocol designs [11, 24, 28], several attempts develop efficient ViTs in PI. For instance, MPCFormer [22] simplifies the exponential in

Softmax by a quadratic function. SALViT [54] and RNAViT [3] employ external attention while MPCViT [51] and PriViT [6] use the ReLU and Square function to replace Softmax.

2.2 Efficient Attention

The quadratic complexity of attention has driven recent efforts to develop linear and Softmax-free attentions. For instance, XNorm [40] and SimA [19] replace Softmax with normalization. Linformer [46] and SOFT [27] use low-rank and decomposed matrices as the kernel function, while CosFormer [37] uses cosine distance. Despite the removal of Softmax, the normalization and matrix multiplication replacements still incur significant overheads in MPC. Specifically, Linformer requires $5.8\times$ and $5.2\times$ more communication and latency overheads compared to Softmax (see Table 1 for details), rendering these approaches still impractical for PI.

2.3 MPC-friendly Softmax Approximations

To bridge the gap between attention and efficient PI, recent efforts have been dedicated to simplifying Softmax with MPC-friendly variants, including 2QuadAttn [22], 2ReLUAttn [31], ScaleAttn [51] and SquareAttn [6]. These approximations typically trade off accuracy for efficiency, among which ScaleAttn is the most efficient but incurs noticeable accuracy loss:

$$\text{ScaleAttn}(Q, K, V) = \frac{1}{N} (QK^T) V, \quad (1)$$

where N is the sequence length. In this paper, we propose a novel Softmax-free attention that is comparably efficient to ScaleAttn while exhibiting the highest accuracy among all approximations.

3 Preliminaries

3.1 Threat Model

Following prior studies [28, 51], we adopt an honest-but-curious threat model in which a client, holding the input data, interacts with a server that owns a pre-trained model to perform private inference. This setting is secure against semi-honest adversaries where they strictly follow the protocols but may attempt to infer more information than allowed.

3.2 Motivations

3.2.1 The intermediate representations are highly correlated across different MHA blocks of ViT. As shown in Figure 2, the attention maps from the CLS token exhibit strong visual and numerical correlations across layers, with the cosine similarity between adjacent layers reaching up to 0.98. This finding provides a feasible opportunity for computing only a subset of the MHA blocks throughout the ViT backbone and reusing their outputs for the remaining blocks, thereby effectively reducing the inference cost of MHA in MPC.

3.2.2 Despite the high correlation, directly copying the attention representations leads to noticeable accuracy drops, necessitating a further trade-off between accuracy and MPC efficiency. While attention reuse can decouple MHA blocks from the ViT architecture, it may simultaneously degrade model capacity [2]. In particular, simply copying the MHA output from layer 2 to subsequent layers 3–5 results in nearly 2% accuracy loss even with retraining. Alternatively, using MHA approximations like SkipAT [43] can partially mitigate

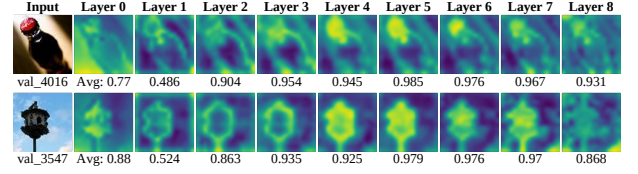


Figure 2: Visualization of attention maps for a 9-depth ViT on Tiny-ImageNet. The value below each attention map of layer ℓ indicates the cosine similarity with that of layer $\ell - 1$.

Table 1: MPC efficiency and accuracy comparison of various attention and activation variants on CIFAR-100. LReLU is the LeakyReLU function. Formal definitions of these attention variants are provided in Appendix A.

Plaintext Linear and Softmax-free Attentions						
Metric	Softmax	Lin	XNorm	SOFT	Cos	SimA
Lat. (s)	1.06	5.48	3.23	0.90	4.16	1.23
Com. (MB)	112.34	653.19	402.58	110.58	278.12	163.71
MPC-friendly Softmax Approximations						
Metric	Softmax	2ReLU	2Quad	Scale	Square	SRAttn
Lat. (s)	1.06	0.54	0.45	0.10	0.29	0.13
Com. (MB)	112.34	97.00	91.58	26.48	29.19	35.51
Acc. (%)	77.49	77.04	76.47	76.56	75.65	77.24
Conventional and Quadratic Activations						
Metric	GeLU	ReLU	LReLU	ELU	Swish	Quad
Lat. (s)	0.82	0.15	0.16	0.36	0.99	0.04
Com. (MB)	103.09	16.00	17.77	49.99	153.07	8.89
Acc. (%)	77.49	76.86	77.13	69.23	76.34	76.42

accuracy loss but incurs $6\times$ more communication overhead in MPC (see Appendix D), rendering them ill-suited for efficient PI.

3.2.3 Different attention and activation variants vary significantly in terms of accuracy and MPC efficiency. Table 1 profiles and compares the accuracy and efficiency of different attention variants in MPC. We observe that ScaleAttn is the fastest variant but incurs a huge accuracy drop, 2ReLUAttn is more accurate but raises the latency. Thus, we combine these two variants and propose the Scale ReLU Attention (SRAttn), achieving a better accuracy-efficiency trade-off:

$$\text{SRAttn}(Q, K, V) = \frac{1}{N} \cdot \text{ReLU}\left(\frac{QK^T}{\sqrt{d_k}}\right) V, \quad (2)$$

where N is the sequence length, d_k is the embedding dimension of each head. Our proposed SRAttn is more MPC-friendly, matching the latency of ScaleAttn while achieving superior accuracy. We also notice that LeakyReLU is more efficient than GeLU with comparable accuracy, and the quadratic function yields the highest efficiency.

3.2.4 Remark. Inspired by these analyses, we identify the following intuitions to develop more MPC-friendly ViTs for efficient PI:

- The dependency across MHA blocks provides an opportunity to reduce MPC computations by reusing attention features, necessitating a careful efficiency-accuracy balance.
- Constructing a ViT with SRAttn and LeakyReLU trades off a marginal accuracy drop for significantly higher MPC efficiency, which forms the foundation of this paper.
- By further developing a mixture of SRAttn and ScaleAttn, it is possible to build more efficient ViTs that achieve comparable performance to the Softmax counterpart.

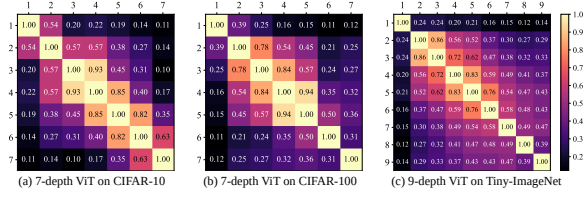


Figure 3: CKA analysis of MHA features on three datasets.

4 MPC-friendly ViT Architecture Co-Design

The overview of SecViT is shown in Figure 4, which integrates three key components: (1) an MPC-aware layer-wise attention reuse and (2) a composite token-wise attention approximation for structurally simplifying Softmax attention, and (3) a sensitivity-aware learnable polynomial replacement for approximating GeLU activation. Collectively, these components address the two main bottlenecks in ViT, bridging accuracy and efficiency.

4.1 MPC-aware Layer-wise Attention Reuse

Driven by the high relevance across MHA blocks, we propose to improve the MPC efficiency by identifying a cluster of highly correlated attention blocks that share similar representation outputs, in which we subsequently reuse the output features of the previous adjacent MHA block instead of computing each MHA in the cluster repeatedly. Our empirical analysis illustrates the existence of such a solution but the determination of reusable MHA cluster and the reusability of MHA representations remain two primary challenges.

4.1.1 Reusable Attention Cluster Determination. To identify the inherent correlation across MHA blocks, we conduct representation analysis based on the Centered Kernel Alignment (CKA) [20] to measure the similarity between each pair of attention output features, where a higher CKA value indicates more correlated representations. CKA is a widely used indicator for correlation evaluation [4, 26] due to its invariance to orthogonal transformations and data scaling. Therefore, the cluster can be formally defined as:

$$C = C \cup \{MHA_\ell \mid CKA(Z_\ell, Z_{\ell-1}) \geq \tau\} \\ = C \cup \left\{ MHA_\ell \mid \frac{\|Z_\ell^T Z_{\ell-1}\|_F^2}{\sqrt{\|Z_\ell^T Z_\ell\|_F^2 \cdot \|Z_{\ell-1}^T Z_{\ell-1}\|_F^2}} \geq \tau \right\}, \quad (3)$$

where Z_ℓ is the representation output of the MHA block at layer ℓ , $\|\cdot\|_F^2$ is the squared Frobenius norm, and τ is the similarity threshold. From Figure 3, we observe that different MHA blocks exhibit varying levels of representation similarity, with intermediate MHA blocks being more highly correlated. This finding suggests a layer-dependent attention reuse strategy to mitigate the feature discrepancies. Note that the CKA evaluation is conducted offline only once and incurs no additional overhead during online inference.

4.1.2 MPC-efficient Attention Representation Adapter. Simply copying the precedent features to the MHA blocks in C incurs a significant accuracy loss. To address this, we propose an MPC-efficient attention representation adapter (ATAdp) denoted as Θ to translate attention features with enhanced translation equivariance for each reusable MHA block independently. We systematically analyze both the accuracy and MPC efficiency of distinct ATAdps composed of

Table 2: Cross-architecture ATAdp comparison on CIFAR-100. FC, G, R and DwC denote fully connected, GeLU, ReLU and depth-wise convolution layers, respectively.

Abbr.	Architecture	Acc.	Com.	A/C
MHA	Standard Attention	77.49	15.64	4.95
LeFF	FC+G+DwC+G+FC	78.23	88.25	0.89
LE	LeFF+ECA	78.60	88.81	0.89
LE _{lite}	LeFF _{lite} +ECA	77.65	18.06	4.30
DRFE	DwC+R+FC+ECA	77.17	8.18	9.43
FRCE	FC+R+FC+ECA	76.70	3.18	24.12
Iden	Identity	75.76	0	—
FE	FC+ECA	75.56	1.31	57.68
ECA	AvgPool+Conv+Sigmoid	75.03	0.56	133.98

locally-enhanced feed-forward network (LeFF) [47], efficient channel attention (ECA) [45] or identity function as shown in Table 2. We observe that (1) the LeFF, LE and LE_{lite} are more accurate than the standard attention but also communication-intensive; (2) the FE and ECA underperform the identity function in both accuracy and communication; (3) the DRFE, FRCE and identity function are more efficient with negligible or little accuracy drop. Through this analysis, we define a candidate ATAdp set that trades off capability and efficiency differently:

$$\Theta_{\text{cap}}(\cdot) = \text{ECA}(\text{FC}(\text{ReLU}(\text{DwC}(\cdot))))), \quad (4)$$

$$\Theta_{\text{lite}}(\cdot) = \text{ECA}(\text{FC}(\text{ReLU}(\text{FC}(\cdot)))), \quad (5)$$

$$\Theta_{\text{eff}}(\cdot) = \text{Identity}(\cdot). \quad (6)$$

More details of selecting candidate ATAdp set are in Appendix D.

To address the irregular similarity across layers, we propose to assign the suitable adapter for each reusable MHA block based on a capability-efficiency trade-off. In principle, higher similarity between an MHA block and its predecessor allows for a more efficient Θ_{eff} to reduce the inference cost, and lower similarity requires a more capable Θ_{cap} to enhance the cross-channel dependencies, while Θ_{lite} is a balanced choice in-between. Concretely, for each ℓ -th MHA block in C , the computation flow with attention reuse is:

$$\hat{Z}_\ell = \begin{cases} \Theta_{\text{cap}}(Z_{\ell-1}) & \text{if } \tau \leq CKA_{\ell,\ell-1} < \tau_c \\ \Theta_{\text{lite}}(Z_{\ell-1}) & \text{if } \tau_c \leq CKA_{\ell,\ell-1} < \tau_t \\ \Theta_{\text{eff}}(Z_{\ell-1}) & \text{if } \tau_t \leq CKA_{\ell,\ell-1} \leq 1 \end{cases} \quad (7)$$

where $CKA_{\ell,\ell-1}$ is the similarity between Z_ℓ and $Z_{\ell-1}$, τ_c and τ_t are the capability-efficiency thresholds for selecting the suitable adapter for each layer.

4.2 Composite Token-wise Attention Approximation

Since the row-wise Softmax is applied independently for each query token, we can assign a cheaper ScaleAttn to unimportant tokens to further reduce the inference cost. To this end, we reformulate the SRAttn as a selective mixture integrated with ScaleAttn:

$$\text{Attn}_\alpha^{\ell,r}(X_r) = \alpha_{\ell,r} \text{SRAttn}(X_r) + (1 - \alpha_{\ell,r}) \text{ScaleAttn}(X_r), \quad (8)$$

where $\alpha_{\ell,r}$ is the architectural selection parameter between two attentions that trade off capability-efficiency differently, for each r -th row of the attention map at layer ℓ . All α s are real-valued and set to 1 at initialization when SRAttn dominates the architecture. Our goal is to minimize the number of SRAttn with more ScaleAttns instead,

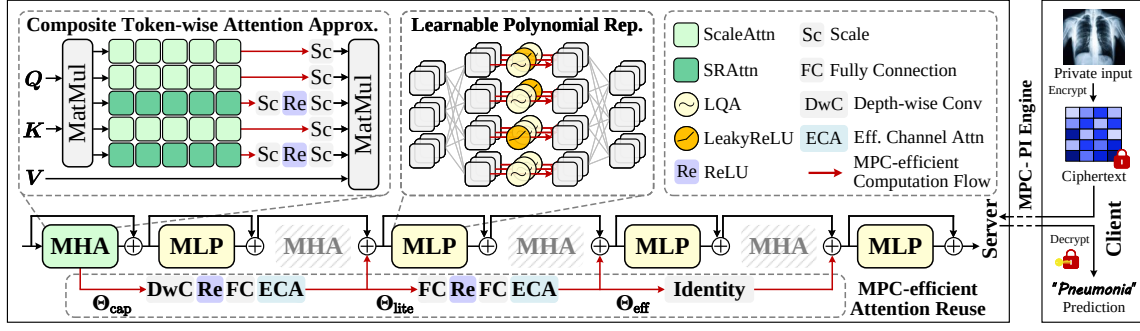


Figure 4: The systematic overview of our proposed SecViT.

through the following ℓ_1 -constrained optimization objective:

$$\mathcal{L}_{\text{Attn}} = \min_{W, \alpha} \mathcal{L}_{\text{CE}}(\mathcal{F}_{W, \alpha}(\mathbf{x}), \mathbf{y}) + \lambda \left(\sum_{\ell=1}^L \sum_{r=1}^N \|\alpha_{\ell, r}\|_1 \right), \quad (9)$$

where $\mathbf{x}$ and $\mathbf{y}$ are the pair of input and label, $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss and λ is the weighted hyperparameter. This training methodology is efficient because we run it only once to obtain a suite of models with different accuracy-efficiency trade-offs by varying the number of SRAttn to be retained. Specifically, after the training is terminated, given the budget of SRAttn denoted as a ratio of B_T , we first identify the $(L \cdot N \cdot B_T)$ -th largest architectural parameter $\hat{\alpha}$ and then re-parameterize all α s as follows:

$$\alpha_{\ell, r} \leftarrow \mathbb{1}[\alpha_{\ell, r} \geq \hat{\alpha}], \quad \forall 1 \leq \ell \leq L, 1 \leq r \leq N, \quad (10)$$

where $\mathbb{1}[\cdot]$ is the indicator function that is 1 if true or 0 otherwise. Finally, we freeze the binarized α in subsequent optimization and perform a distillation-based fine-tuning with the baseline model as the teacher to recover accuracy.

4.3 Sensitivity-aware Polynomial Replacement

In this section, we propose a sensitivity-driven loss-centric metric to assess the activation importance and replace ineffectual ones with learnable approximants to further improve efficiency while maintaining high accuracy.

4.3.1 Activation Sensitivity Quantification. To bridge the activation and loss function, we define a parametric activation as:

$$\sigma_{\beta}^{\kappa}(x) = \beta_{\kappa} \cdot \sigma(x) + (1 - \beta_{\kappa}) \cdot x, \quad (11)$$

where $\sigma(\cdot)$ is the LeakyReLU function and β_{κ} is the switch parameter of the κ -th activation. Similar to α , all β s are initialized to 1 without affecting the model functionality. We set $\beta_{\kappa, -\kappa} = [0, \dots, 0, \beta_{\kappa}, 0, \dots, 0]$ and formally define the sensitivity of the κ -th activation as the magnitude of the gradient-parameter product:

$$S_{\kappa} = |\beta_{\kappa, -\kappa}^{\top} \nabla_{\beta} \mathcal{L}(\beta)|. \quad (12)$$

The definition of S_{κ} is a derivation of the first-order Taylor expansion of $\mathcal{L}(\cdot)$ with respect to β_{κ} at β , approximating the absolute variation of the loss given the removal of β_{κ} :

$$\beta_{\kappa, -\kappa}^{\top} \nabla_{\beta} \mathcal{L}(\beta) \approx \lim_{\xi \rightarrow 0} \frac{\mathcal{L}(\beta) - \mathcal{L}(\beta - \xi \beta_{\kappa, -\kappa})}{\xi} \Big|_{\beta=1}. \quad (13)$$

The activations with high sensitivity are of significant effect on the loss and should be kept, while insensitive ones can be replaced with efficient alternatives to reduce computational overhead. To stabilize

the training dynamics, we introduce the knowledge distillation (KD) [13] loss as a regularization during the sensitivity evaluation:

$$\mathcal{L}_{\text{sensv}} = \lambda_1 \mathcal{L}_{\text{CE}}(\mathcal{F}_{W, \beta}(\mathbf{x}), \mathbf{y}) + \lambda_2 \mathcal{L}_{\text{KL}}(\mathcal{F}_W(\mathbf{x}), \mathcal{F}_{W, \beta}(\mathbf{x})), \quad (14)$$

where $\mathcal{L}_{\text{KL}}$ is the KD loss and λ_1, λ_2 are hyperparameters. The combination of $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_{\text{KL}}$ ensures that the removal of each activation minimally perturbs both the supervised loss and prediction discrepancy. To decide the final activation configuration asymptotically, we use a gradually increased rather than invariant sparsity throughout the E_S epochs by a cubically decreasing scheduler [25, 39]:

$$B_C^{(e)} = \begin{cases} B_C^i & , \quad 0 \leq e < e_i \\ B_C + (B_C^i - B_C) \left(1 - \frac{e - e_i}{e_f - e_i}\right)^3 & , \quad e_i \leq e < e_f \\ B_C & , \quad e_f \leq e < E_S \end{cases} \quad (15)$$

where the scheduled budget at epoch e , $B_C^{(e)}$, is decreased from an initial value B_C^i (typically K) to a final budget B_C over $e_f - e_i$ iteration steps, preventing drastic model degradation in early sparsification. Concurrently, we generate the mask associated with $B_C^{(e)}$ for each step e using the top- k rule where K is the number of activations:

$$\mathcal{M}_{\kappa}^{(e)} = \begin{cases} 1 & , \quad \text{if } S_{\kappa} \text{ in top-} B_C^{(e)} \text{ of } \{S_j\}_{j=1}^K \\ 0 & , \quad \text{otherwise} \end{cases} \quad (16)$$

The mask is applied by $\beta \leftarrow \beta \odot \mathcal{M}^{(e)}$ and updated at each step until when we freeze $\mathcal{M}$ at the end of training.

4.3.2 Learnable Quadratic Approximant Replacement. As observed in Table 1, the Quad function is efficient in MPC but suffers from accuracy loss. Therefore, we propose to retain LeakyReLU where $\mathcal{M}_{\kappa} = 1$, and use an element-wise learnable quadratic approximant (LQA) as an alternative to the non-linear activation where $\mathcal{M}_{\kappa} = 0$:

$$\text{LQA}(x) = \epsilon \cdot c_1 x^2 + c_2 x + c_3, \quad (17)$$

where c_1, c_2, c_3 are trainable coefficients that are initialized to 0, 1, 0, respectively. We also add a small constant ϵ prior to c_1 to prevent gradient explosion, analogous to [35, 36]. Therefore, the activation layers of $\mathcal{F}_{W, c}$ can be rewritten as follows:

$$\hat{\sigma}(x) = \mathcal{M} \cdot \sigma(x) + (1 - \mathcal{M}) \cdot \text{LQA}(x). \quad (18)$$

To facilitate the polynomial learning, we introduce a layer-wise feature-aware activation consistency (FAC) constraint to mitigate intermediate prediction discrepancies:

$$\mathcal{L}_{\text{LQA}} = \lambda_1 \mathcal{L}_{\text{CE}} + \lambda_2 \mathcal{L}_{\text{KL}} + \lambda_3 \sum_{\ell=1}^L \text{MSE} \left(\frac{A_T^{\ell}}{\|A_T^{\ell}\|_p}, \frac{A_S^{\ell}}{\|A_S^{\ell}\|_p} \right), \quad (19)$$

Table 3: Performance and MPC efficiency comparison under LAN and WAN settings on different datasets.

Approaches	Com.	CIFAR-10			Acc.	Com.	CIFAR-100			Acc.	Com.	Tiny-ImageNet			Acc.
		LAN	WAN				LAN	WAN				LAN	WAN		
Baseline ViT	254.32	6.99	119.46	94.56		254.49	7.00	119.50	77.49		5885.98	167.15	1527.82	64.45	
MPCFormer	139.36	3.89	68.90	93.65		139.54	3.90	68.92	76.66		4275.10	123.49	1134.63	61.43	
SALViT	223.65	6.11	91.81	94.35		223.84	91.82	223.83	74.88		1895.44	53.37	515.20	60.54	
	225.30	6.16	92.59	95.23		225.48	92.55	225.48	76.40		2237.79	63.01	596.48	61.39	
	226.96	6.19	93.31	95.02		227.13	93.37	227.13	76.65		2580.15	73.11	688.46	62.22	
RNaviT	141.41	4.00	68.50	93.06		141.59	4.00	68.47	74.48		789.00	22.79	230.49	62.73	
MPCViT	176.82	4.94	78.47	93.59		177.00	4.95	79.93	76.40		1190.95	33.68	337.24	62.65	
	191.74	5.33	85.42	94.08		191.92	5.32	85.42	76.93		2045.96	58.05	548.25	63.38	
	204.17	5.68	89.10	94.22		204.35	5.64	89.08	76.92		2862.11	80.72	754.53	63.45	
	219.09	6.00	93.33	94.27		219.26	6.03	94.63	77.46		3717.13	105.68	986.31	63.03	
PriViT	86.09	2.94	93.51	90.04		86.27	2.92	94.91	70.00		1054.33	34.78	334.82	60.23	
	123.16	3.72	95.72	92.47		123.34	3.73	95.75	74.91		2137.28	64.43	602.10	62.18	
	204.23	4.59	101.45	93.74		204.40	4.62	101.41	75.85		3199.41	94.35	857.92	63.66	
	197.29	5.62	108.03	94.11		197.47	5.62	108.06	76.57		4261.54	123.85	1116.02	64.26	
SecViT (ours)	79.85	2.52	56.08	94.02		80.03	2.59	55.63	77.06		375.83	14.48	143.52	61.93	
	83.20	2.65	56.57	94.61		83.38	2.67	56.11	78.21		480.09	17.35	165.67	63.41	
	85.62	2.72	56.38	94.80		85.80	2.75	56.40	78.77		585.28	20.53	196.84	64.44	
	87.40	2.78	57.23	95.10		87.58	2.85	57.05	78.61		619.54	21.70	207.34	65.15	

where λ_3 is the weight of the FAC loss, $\mathcal{A}_T^\ell$ and $\mathcal{A}_S^\ell$ are the pair of teacher and student activation maps at layer ℓ . Similar to [35, 49], we set $p = 2$ to calculate the normalized ℓ_2 feature distance. The FAC loss forces the polynomial model to mimic the hidden features of the baseline across different layers and determines more accurate coefficients of LQA, yielding superior model performance. The pseudocode of SecViT is provided in Appendix B.

5 Experiments

5.1 Experimental Setup

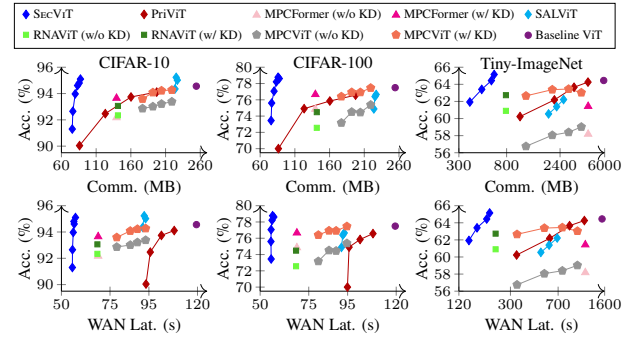
5.1.1 Testbed Environment. SecViT is built on top of the SecretFlow-SPU library (v0.9.3b0) [29] for MPC-based PI. Following [51], we simulate the LAN and WAN network settings via Linux Traffic Control, where the bandwidth and the round-trip time are (377Mbps, 0.3ms) and (40Mbps, 80ms), respectively.

5.1.2 Architectures and Datasets. We evaluate SecViT on different ViT variants across three datasets. Specifically, we set the depth, the number of heads and the hidden dimension to $\{7, 4, 256\}$ for CIFAR-10/100, and $\{9, 12, 192\}$ for Tiny-ImageNet, respectively. Note that we use the ViT-lite architecture [12] to reproduce all competitive results under the same network setting for a consistent comparison.

5.1.3 Concrete Configurations. To select the suitable adapter for each layer, we set thresholds $\tau_r, \tau_c, \tau_t = 0.7, 0.8, 0.9$, and set (B_T, B_C) to $\{(0.1, 0.1), (0.3, 0.3), (0.5, 0.5), (0.7, 0.7), (0.9, 0.9)\}$ as default settings to trade off efficiency and accuracy. We provide further details of the experimental settings and hyperparameters in Appendix C.

5.2 Performance Evaluation and Comparison

5.2.1 Comparison with Network-level Methods. The results in Table 3 show that SecViT outperforms prior network-level methods in both LAN and WAN settings. Specifically, on CIFAR-100, SecViT achieves 2.11% accuracy boost with 1.6× communication

**Figure 5: Pareto comparison with prior works.**

and 1.2× latency reductions over MPCFormer. Moreover, SecViT exhibits 10.1×, 7.3×, 6.0× communication savings and 7.8×, 5.7×, 4.6× latency reductions with iso-accuracy compared to the baseline, PriViT and MPCViT, respectively, on Tiny-ImageNet. Additionally, as shown in Figure 5, SecViT is Pareto dominant across all evaluated datasets. The consistent performance gains and efficiency improvements validate the effectiveness of SecViT.

5.2.2 Comparison with SOTA Protocol-level Methods. We further compare SecViT with SOTA PI protocols including Nimbus [24] and BumbleBee [28]. Note that both methods evaluate linear layers with HE which is slower than MPC. To keep a fair comparison, we adopt their non-linear layer implementations while keeping linear layers consistent with SecViT. As shown in Table 4, SecViT consistently surpasses Nimbus and BumbleBee across three datasets and achieves up to 9.7× and 10.0× communication savings as well as 7.9× and 8.2× latency reductions under a LAN setting on Tiny-ImageNet, respectively, validating that SecViT is compatible with existing PI protocols and can be combined for further benefits.

5.2.3 Micro Performance Breakdown. We profile the layer-wise communication breakdown of attention and activation operations in Figure 6. For attention optimization, SecViT achieves the lowest

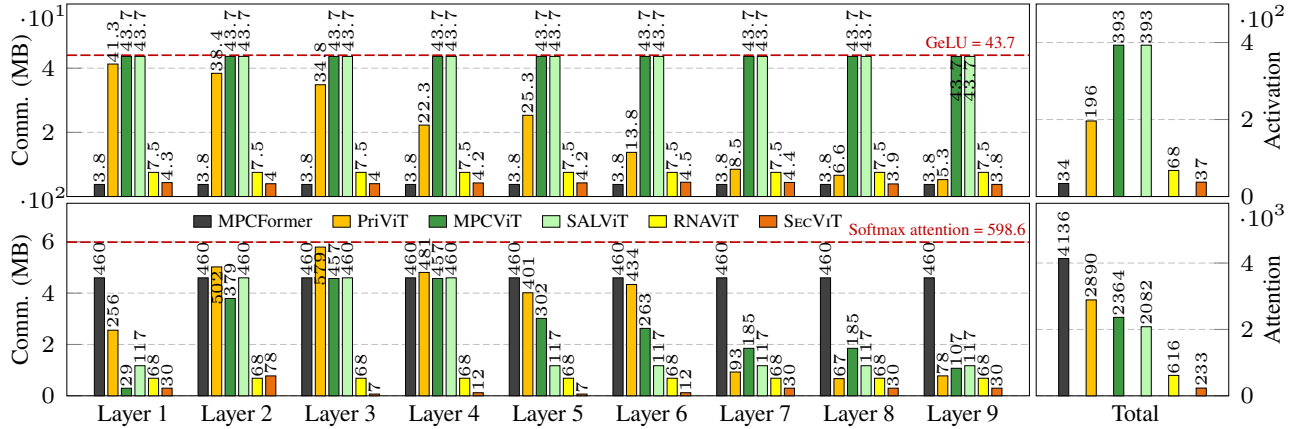


Figure 6: Layer-wise breakdown of communication overheads for attention and activation operations on Tiny-ImageNet.

Table 4: Comparison with protocol-level approaches.

Dataset	Method	Nimbus			BumbleBee		
		Com.	LAN	WAN	Com.	LAN	WAN
C10	Baseline	191.1	5.4	99.1	241.0	6.7	114.8
	+SECViT	81.4	2.6	56.4	86.8	2.8	57.1
C100	Baseline	191.3	5.4	99.1	241.2	6.7	114.8
	+SECViT	81.6	2.6	55.9	86.9	2.8	57.4
Tiny	Baseline	5592.5	159.7	1473.9	6262.6	177.6	1652.5
	+SECViT	578.6	20.2	198.4	626.4	21.7	208.8

communication across almost all layers, especially for the reused layers from 3 to 6. For activation optimization, MPCFormer yields the lowest communication by replacing all GeLUs with a static Quad function $\text{Quad}(x) = 0.125x^2 + 0.25x + 0.5$, incurring significant accuracy loss. In contrast, SECViT develops a learnable quadratic approximant to maintain the model performance with a slightly increased communication (37MB vs. 34MB), which is considered acceptable. The results demonstrate the effectiveness of our structural optimizations for both attention and activation operations.

5.2.4 Layer-wise Attention Operation Analysis. We also analyze the detailed layer-wise breakdown of attention configurations of SECViT and other competitive approaches in Table 6. Note that different approaches approximate Softmax at varying granularities (layer, head, token), we count the number of most fine-grained token-wise attention before the Softmax function (or its approximation) for a consistent comparison. It can be observed that the optimized communications of MPCFormer, SALViT, MPCViT and PriViT still remain at a high level. RNAViT achieves relatively low communication cost by using two additional matrices to compress the dimension of QK^T , thus reducing the number of Softmax operations compared to other approximations but incurring a noticeable accuracy drop. Our SECViT uses efficient attention representation adapters to reuse the features of layer 2 across subsequent layers from 3 to 6, avoiding costly attention computations in MPC and achieving the lowest communication with negligible accuracy loss.

5.3 Ablation Studies of SECViT

5.3.1 Effect of ATAdp Configurations. In Table 5, we study the effect of ATAdp with different configurations and layer spans. We

Table 5: Impact of different combinations of ATAdp.

ATAdp	Layers	Acc.	Com.	WAN
–	–	64.35	665.1	206.1
cap	{3, 4, 5, 6}	64.37	595.7	199.9
lite	{3, 4, 5, 6}	63.89	574.8	197.5
eff	{3, 4, 5, 6}	62.75	547.9	179.4
{cap, lite, eff}	{3, 5, 7, 9}	63.28	590.5	201.0
{cap, lite, eff}	{3, 4, 5, 6, 7, 8, 9}	61.77	397.0	150.9
{cap, lite, eff}	{3, 4, 5, 6}	64.44	585.3	196.8

observe that expanding the span of ATAdp or using only the identity function leads to a significant accuracy drop. The default setting outperforms all others in accuracy and is also comparably efficient.

5.3.2 Effect of Different Polynomial Approximants. In Figure 7, we evaluate the impact of activation approximants with different polynomial terms and degrees. It can be concluded that LQA is a learnable version of Quad and provides improved accuracy with identical communication cost, while the higher third-order polynomial, i.e., LCA, yields little accuracy boost but is nearly as expensive as LeakyReLU. Therefore, LQA is a more preferred and MPC-efficient approximant that trades off better accuracy and efficiency.

5.3.3 Effectiveness of Losses in Polynomial Replacement. We also verify the importance of each loss for determining LQA coefficients. As shown in Table 7, both $\mathcal{L}_{KL}$ and $\mathcal{L}_{FAC}$ benefit the model performance, while $\mathcal{L}_{FAC}$ improves more accuracy than $\mathcal{L}_{KL}$ relative to using neither of them. More importantly, combining both yields the optimal accuracy, indicating that all losses are indispensable.

5.3.4 Effectiveness of Different Optimizations. In Figure 8, we validate the effectiveness of each optimization by applying them step by step to a 9-depth ViT on Tiny-ImageNet. We observe that directly replacing the Softmax attention and GeLU with SRAttn and LeakyReLU incurs a noticeable accuracy drop. Our attention reuse technique reduces the inference cost and significantly recovers model accuracy. The LQA function with learned coefficients is an ideal alternative that reduces communication by 10.1× and latency by 7.8× with negligible accuracy loss compared to the baseline. The results demonstrate the necessity of our architectural designs for building performant and MPC-efficient ViTs.

Table 6: Layer-wise attention operation breakdown of a 9-depth ViT on Tiny-ImageNet. Communication is reported in MB. EA and SA are the short for external and Softmax attention with L2Q approximation used in [54]. The total communication is not strictly equal to the sum of each layer due to rounding bias.

Layer	Baseline			MPCFormer			SALViT			RNAViT			MPCViT			PriViT			SecViT		
	Attn.	#Opt.	Com.	Attn.	#Opt.	Com.	Attn.	#Opt.	Com.	Attn.	#Opt.	Com.	Attn.	#Opt.	Com.	Attn.	#Opt.	Com.	Attn.	#Opt.	Com.
1	Softmax	3084	598.6	2Quad	3084	459.5	EA	3084	117.2	RNA	1536	68.4	2ReLU Scale	0	0	Softmax	1164	225.9	SRAttn Scale	12	0.3
																					29.3
2	Softmax	3084	598.6	2Quad	3084	459.5	SA	3084	459.5	RNA	1536	68.4	2ReLU Scale	2313	371.9	Softmax	2544	493.8	SRAttn Scale	2724	74.1
																					3.4
3	Softmax	3084	598.6	2Quad	3084	459.5	SA	3084	459.5	RNA	1536	68.4	2ReLU Scale	2827	454.5	Softmax	2976	577.7	Θ_{lite}	—	6.7
4	Softmax	3084	598.6	2Quad	3084	459.5	SA	3084	459.5	RNA	1536	68.4	2ReLU Scale	2827	454.5	Softmax	2424	470.5	Θ_{cap}	—	12.0
5	Softmax	3084	598.6	2Quad	3084	459.5	EA	3084	117.2	RNA	1536	68.4	2ReLU Scale	1799	289.2	Softmax	1980	384.3	Θ_{lite}	—	6.7
6	Softmax	3084	598.6	2Quad	3084	459.5	EA	3084	117.2	RNA	1536	68.4	2ReLU Scale	1542	247.9	Softmax	2160	419.3	Θ_{cap}	—	12.0
7	Softmax	3084	598.6	2Quad	3084	459.5	EA	3084	117.2	RNA	1536	68.4	2ReLU Scale	1028	165.3	Softmax	252	48.9	SRAttn	12	0.3
																					29.3
8	Softmax	3084	598.6	2Quad	3084	459.5	EA	3084	117.2	RNA	1536	68.4	2ReLU Scale	1028	165.3	Softmax	108	21.0	SRAttn	12	0.3
																					29.3
9	Softmax	3084	598.6	2Quad	3084	459.5	EA	3084	117.2	RNA	1536	68.4	2ReLU Scale	514	82.6	Softmax	168	32.6	SRAttn	12	0.3
																					29.3
Total	—	—	5387.6	—	—	4135.9	—	—	2081.8	—	—	615.9	—	—	2363.8	—	—	2889.8	—	—	233.5

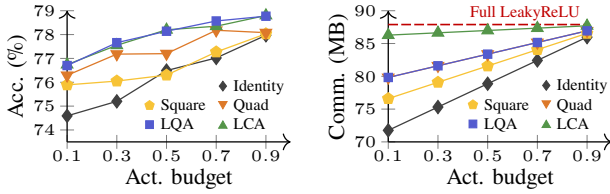


Figure 7: Impact of different activation replacements on CIFAR-100. LQA and LCA are short for learnable quadratic and cubic approximants, respectively.

Table 7: Impact of different combinations of $\mathcal{L}_{LQA}$.

Loss		Budget Spectrum: (B_r, B_c)			
$\mathcal{L}_{KL}$	$\mathcal{L}_{FAC}$	(0.1, 0.1)	(0.3, 0.3)	(0.5, 0.5)	(0.7, 0.7)
✗	✗	69.17	72.42	74.93	76.44
✓	✗	72.82	74.33	76.49	77.13
✗	✓	73.39	75.66	77.20	77.92
✓	✓	75.60	77.06	78.21	78.77

5.3.5 Thresholds Sensitivity. To investigate how τ , τ_c , τ_t impact the performance of attention reuse and the associated MPC efficiency, we conduct SecViT training with different values of $\{\tau, \tau_c, \tau_t\}$. As shown in Table 8, we find that smaller thresholds usually expand the span of C and use more identity functions to approximate MHA blocks, incurring a significant accuracy drop. Conversely, raising the thresholds fails to fully exploit the attention correlations and leads to diminishing returns in accuracy improvements over the default setting $\tau = 0.7$, $\tau_c = 0.8$, $\tau_t = 0.9$. Therefore, we use $\tau = 0.7$, $\tau_c = 0.8$, $\tau_t = 0.9$ unless otherwise stated.

6 Conclusion and Future Work

In this paper, we address the gap between the architectural properties of ViT and MPC-based PI, and propose SecViT, an MPC-efficient inference framework that adapts ViT models to privacy-friendly counterparts and achieves comparable performance while

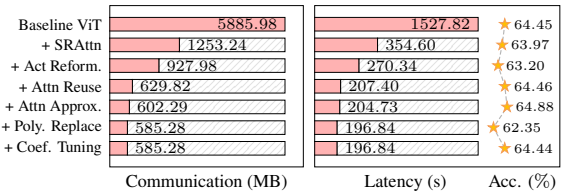


Figure 8: Ablation study of each optimization in SecViT.

Table 8: Ablation study of different thresholds τ , τ_c , τ_t .

τ	τ_c	τ_t	C	Θ	Acc.	Com.
0.3	0.5	0.7	{3, 4, 5, 6, 7, 8, 9}	{eff, eff, eff, eff, lite, cap, cap}	61.57	354.46
0.5	0.6	0.7	{3, 4, 5, 6, 7}	{eff, eff, eff, eff, cap}	62.08	503.54
0.5	0.7	0.9	{3, 4, 5, 6, 7}	{lite, lite, lite, lite, cap}	63.40	530.43
0.6	0.7	0.8	{3, 4, 5, 6}	{eff, lite, eff, lite}	63.82	561.38
0.8	0.9	0.95	{3, 5}	{cap, cap}	64.42	630.74
0.9	0.95	0.99	$\emptyset$	$\emptyset$	64.35	665.12
0.7	0.8	0.9	{3, 4, 5, 6}	{lite, cap, lite, cap}	64.44	585.28

significantly reducing both communication and latency overheads. SecViT employs a mixture of attention that tailors optimal configurations across different layers and tokens, and develops a sensitivity-aware differentiable activation replacement to further improve inference efficiency. Extensive experiments demonstrate the superiority of SecViT over both network and protocol-level approaches, and validate the effectiveness of our key designs. For future work, we believe that SecViT is generalizable and can be extended to more Transformer-based architectures such as large language models (LLMs) and vision language models (VLMs).

Acknowledgments

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A Formal Definitions of Attention Variants

Several approximations have been proposed to mitigate the cost of Softmax attention in plaintext and private Transformer inference. In this section, we briefly describe their formal definitions.

Linformer [46] uses a low-rank matrix to approximate the self-attention while keeping the Softmax function:

$$\text{Linformer}(Q, K, V) = \text{Softmax}\left(\frac{Q(EK)^T}{\sqrt{d_k}}\right) \cdot (FV), \quad (20)$$

where $E, F \in \mathbb{R}^{N \times k}$ are two matrices that project the original $(N \times d_k)$ -dimensional K, V into $(k \times d_k)$.

XNorm is proposed in UFO-ViT [40] to reduce the self-attention from quadratic to linear complexity by replacing Softmax with normalization:

$$\begin{aligned} \text{XNorm}(Q, K, V) &= \text{XN}_{\text{dim}_1}(Q) (\text{XN}_{\text{dim}_2}(K^T V)), \\ \text{XN}(x) &:= \frac{\gamma x}{\sqrt{\sum \|x\|^2}}, \end{aligned} \quad (21)$$

where γ is a learnable parameter.

SOFT [27] is also a linear attention that approximates Softmax using normalization and $\exp(\cdot)$ operations:

$$\text{SOFTAttn}(Q, K, V) = \exp\left(-\frac{\|Q - K\|_2^2}{2\sqrt{d_k}}\right) V. \quad (22)$$

cosFormer [37] proposes to satisfy the non-negativity and non-linear re-weighting properties of Softmax attention via a ReLU and cosine-based distance re-weighting function:

$$\text{cosFormer}(Q, K, V) = \cos(\text{ReLU}(Q), \text{ReLU}(K)) V, \quad (23)$$

where $\cos$ is the cosine-based re-weighting function.

SimA [19] uses normalized query and key matrices with simple ℓ_1 -norm instead of using Softmax:

$$\text{SimA}(Q, K, V) = \frac{Q}{\|Q\|_1} \left(\frac{K}{\|K\|_1} \right)^T V. \quad (24)$$

We observe that such “plaintext-domain” linear attentions heavily rely on normalization or low-rank matrix to approximate the Softmax function. These solutions remain ill-suited for PI because the matrix multiplications, comparison, and reciprocal involved still incur substantial overheads in MPC. To alleviate this, several MPC-optimized attention variants are proposed, including:

2Quad Attention is proposed in MPCFormer [22] by replacing exponential with square function:

$$2\text{QuadAttn}(Q, K, V) = \frac{\left(\frac{QK^T}{\sqrt{d_k}} + \mu\right)^2}{\sum_{i=1}^N \left(\frac{QK^T}{\sqrt{d_k}} + \mu\right)_i^2} \cdot V, \quad (25)$$

where μ is a small value to avoid a zero denominator.

2ReLU Attention is proposed in SecureML [31] by replacing exponential with ReLU function:

$$\begin{aligned} 2\text{ReLUAttn}(Q, K, V) &= \text{ReLU}\text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V, \\ \text{where ReLU}\text{Softmax} &= \frac{\text{ReLU}(x_i)}{\sum_i \text{ReLU}(x_i) + \mu} \end{aligned} \quad (26)$$

Scale Attention is proposed in MPCViT [51] and can be simply formulated as:

$$\text{ScaleAttn}(Q, K, V) = \frac{(QK^T)}{N} V = \frac{Q}{\sqrt{N}} \left(\frac{K^T}{\sqrt{N}} \right) V \quad (27)$$

where N is the token length.

Square Attention is proposed in PriViT [6] by replacing exponential with simplified square function:

$$\text{SquareAttn}(Q, K, V) = \frac{(QK^T)^2}{N} V \quad (28)$$

B Algorithmic Details of SECViT

In this section, we provide the algorithmic procedure of SECViT in Algorithm 1. The training pipeline is divided into three primary stages: (1) MPC-aware layer-wise attention reuse (line 1-3), (2) composite token-wise attention approximation (line 4-11), and (3) sensitivity-aware polynomial replacement (line 12-20).

C Implementation Details

In this section, we provide more details of the 2PC library, datasets and model architectures used in our experiments. We also report the baseline results, MPC efficiency and specific hyperparameters.

C.1 Baseline Architectures

Note that baseline architectures vary differently across prior works. For instance, MPCFormer [22] focuses on BERT. SALViT [54] and RNAViT [3] use compact convolution transformer (CCT) [12] while PriViT [6] uses ViT-Tiny [41]. To keep a consistent comparison, following prior approaches [48, 51], we use ViT-lite from [12] to

Algorithm 1 Pseudocode of SecViT for developing MPC-friendly ViTs.

Input: $\mathcal{F}_W$: Baseline Softmax-GeLU ViT, $\mathcal{F}_W^{\text{SL}}$: Pre-trained SRAttn-LeakyReLU ViT, τ, τ_c, τ_t : Similarity thresholds, B_T : Attention budget, B_C : Activation budget, E_T, E_C : Training epochs.

Output: $\mathcal{F}_W^{\text{MPC}}$: Performant and MPC-efficient ViT with mixture of attention and activation.

- 1: Perform Centered Kernel Alignment analysis on $\mathcal{F}_W^{\text{SL}}$ and obtain the reusable attention cluster C according to Eq. 3.
- 2: Apply attention adapter $\{\Theta_{\text{eff}}, \Theta_{\text{lite}}, \Theta_{\text{cap}}\}$ to each MHA block in C according to Eq. 7.
- 3: Fine-tune $\mathcal{F}_W^{\text{SL}}$ distilled by the teacher $\mathcal{F}_W$.
- 4: Set α active while β frozen.
- 5: **for** each epoch e_t in E_T **do**
- 6: Update (W, α) using AdamW optimizer according to loss $\mathcal{L}_{\text{Attn}}$ in Eq. 9.
- 7: Adjust learning rate using the cosine scheduler.
- 8: **end for**
- 9: $\hat{\alpha} \leftarrow \min\{\text{Top-K}(\alpha, \text{floor}(L \cdot N \cdot B_T))\}$.
- 10: Re-parameterize and freeze $\alpha \leftarrow \mathbb{1}[\alpha \geq \hat{\alpha}]$.
- 11: Fine-tune $\mathcal{F}_W^{\text{SL}}$ distilled by the teacher $\mathcal{F}_W$, set β active.
- 12: **for** each epoch e_c in E_C **do**
- 13: Calculate the importance score S_K for all β according to Eq. 12, and $B_C^{(e)}$ according to Eq. 15.
- 14: $\hat{S} \leftarrow \min\{\text{Top-K}(\{S_K\}_{K=1}^K, \text{floor}(B_C^{(e)} \cdot K))\}$.
- 15: Update mask $M \leftarrow \mathbb{1}[S \geq \hat{S}]$ and $\beta \leftarrow \beta \odot M$.
- 16: **end for**
- 17: Reformulate $\mathcal{F}_{W,c}^{\text{LQA}} \leftarrow \mathcal{F}_W^{\text{SL}}$ according to Eq. 18 and freeze β .
- 18: **while** not converged **do**
- 19: Update (W, c) using AdamW optimizer according to loss $\mathcal{L}_{\text{LQA}}$ in Eq. 19.
- 20: **end while**
- 21: **return** The final converged MPC-efficient ViT $\mathcal{F}_W^{\text{MPC}}$.

implement the baseline architectures and reproduce all the accuracy and efficiency results of competitive methods using the same baselines in our experiments, as detailed in Table 9.

C.2 Training Details and Hyperparameters

We provide the detailed training settings in Table 10, including our attention reuse policies, epochs of each training phase and specific hyperparameters.

D Details of Deciding Candidate ATAdp Set

In this section, we discuss the trade-offs between capability and efficiency of different adapter functions, and elucidate the underlying rationales that guide the selection of adapters in this work. Let $\Theta : \mathbb{R}^{N \times d} \rightarrow \mathbb{R}^{N \times d}$ be the attention representation adapter that transforms the output of MHA block from layer $\ell - 1$ to ℓ as $\hat{Z}_\ell = \Theta(Z_{\ell-1})$. Notably, $\hat{Z}_\ell$ can be either a direct replica or an approximation of $Z_{\ell-1}$, depending on whether Θ is an identity or another encoding function, including:

Locally-enhanced Feed-forward Network (LeFF) is proposed in [47] to enhance the local context encoding capability of standard MLP block by adding a depth-wise convolution (DwC) between two fully connected (FC) layers interleaved with GeLU functions:

$$\begin{aligned}
 \text{LeFF}(x) &= \text{FC}_2(\text{DwC}(\text{FC}_1(x))), \\
 \text{FC}_1 &= \text{GeLU}(\text{Linear}(x)), \\
 \text{DwC} &= \text{GeLU}(\text{Conv}(x)), \\
 \text{FC}_2 &= \text{Linear}(x).
 \end{aligned} \tag{29}$$

Efficient Channel Attention (ECA) is a 1D convolution-based attention module [45] designed to capture local cross-channel interactions and can be formulated as:

$$\text{ECA}(x) = \text{Sigmoid}(\text{Conv1D}(\text{AvgPool}(x))) \tag{30}$$

Note that the existing plaintext-domain method SkipAT [43] uses both LeFF and ECA to approximate the MHA blocks and argue that such a combination is computationally simpler and faster compared to the standard Softmax attention. However, **such an assumption is no longer valid for MPC-based inference**. We breakdown the communication of each component within the LeFF and ECA in Table 11, and observe that despite the higher accuracy, LeFF+ECA requires 88.81 MB in total, which is 5.5× more than that of MHA. To identify more efficient candidate adapter functions, we evaluate an extensive range of architectural combinations for Θ by changing the kernel size of DwC and the channel expansion of FC, and replacing GeLU with ReLU as shown in Table 12. From the comparison, we observe that: (1) DwC is the most communication-intensive component and using smaller kernel size (3×3) can substantially reduce the associated overhead; (2) replacing GeLU with ReLU trades off a marginal accuracy drop for significantly improved MPC efficiency; (3) ECA module exhibits high MPC efficiency, yet deploying it independently incurs a considerable accuracy loss; (4) the identity function is more accurate and efficient than Le_c and ECA, but should be carefully used for MHA blocks with high similarity, preventing a significant drop in model capability. Through a thorough evaluation, we select $\Theta_{\text{cap}}, \Theta_{\text{lite}}, \Theta_{\text{eff}}$ as the candidate attention adapters according to the following principles:

Table 9: The detailed architectures of baseline ViTs used in our experiments.

Dataset	Model	Architecture					Param. (M)	MACs (M)	Acc. (%)	Com. (MB)	Latency (s)	
		Depth	Dim.	# Head	MLP Dim.	Patch					LAN	WAN
CIFAR-10	ViT	7	256	4	512	4×4	3.72	240.34	94.56	254.32	6.99	119.46
CIFAR-100	ViT	7	256	4	512	4×4	3.74	240.36	77.49	254.49	7.00	119.50
Tiny-ImageNet	ViT	9	192	12	384	4×4	2.77	688.28	64.45	5885.98	167.15	1527.82

Table 10: Training details and experimental settings in SecViT. We use the AdamW optimizer and cosine scheduler for all the training. Note that AR, AA, PR, PC are the abbreviations of attention reuse, attention approximation, polynomial replacement, polynomial coefficient training, respectively.

Dataset	Model	C	Θ	Epochs				Batch Size	Learning Rate	Hyperparameters			
				AR	AA	PR	PC			λ	λ_1	λ_2	λ_3
CIFAR-10	ViT ₇₋₄₋₂₅₆	{4, 5, 6}	{eff, lite, lite}	300	300	120	300	128	55e-5	2e-4	0.1	0.9	100
CIFAR-100	ViT ₇₋₄₋₂₅₆	{3, 4, 5}	{cap, lite, eff}	300	300	120	300	128	6e-4	2e-4	0.1	0.9	100
Tiny-ImageNet	ViT ₉₋₁₂₋₁₉₂	{3, 4, 5, 6}	{lite, cap, lite, cap}	180	180	90	180	64	5e-4	2e-4	0.1	0.9	50

Table 11: Atomic operations and communication comparison between standard MHA and LeFF+ECA. We observe that, despite the higher accuracy, the architecture of “LeFF+ECA” incurs $5.5 \times$ more communication than the standard attention due to the expensive depth-wise convolution operation with 5×5 kernel size, thus remaining an impractical alternative for efficient PI.

Arch.	Atomic operation and communication with an input dimension of (1, 65, 256)							Total Com.	Acc.	
MHA	$Q, K, V = XW^Q, XW^K, XW^V$ 2.01 MB			QK 0.38 MB	$\sqrt{d_k}$ 0.13 MB	Softmax 12.27 MB	QKV 0.38 MB	Proj. 0.88 MB	16.05	77.49
LeFF+ECA	Locally-enhanced Feed-forward Network (LeFF)					Efficient Channel Attention (ECA)			88.81	78.60
	FC (2×) 1.38 MB	GeLU 14.5 MB	DwC (5 × 5) 56.5 MB	GeLU 14.5 MB	FC 1.38 MB	AvgPool 0.0019 MB	Conv & Transpose 0.39 MB	Sigmoid 0.16 MB		

Table 12: Cross-architecture comparison of attention adapter. By varying the architectural hyperparameters and combining distinct components step by step, we obtain a series of ATAdp variants that trade off accuracy and efficiency differently.

Arch.	Atomic Operations								Total Com.	Acc.
	Locally-enhanced Feed-forward Network (LeFF)				Efficient Channel Attention (ECA)					
LeFF	FC (2×)	GeLU	DwC (5 × 5)	GeLU	FC	–	–	–	88.25	78.23
LE ₁	FC (1.5×)	ReLU	DwC (3 × 3)	ReLU	FC	AvgPool	Conv & Transpose	Sigmoid	18.06	77.65
LE ₂	FC (1.5×)	–	DwC (3 × 3)	ReLU	FC	AvgPool	Conv & Transpose	Sigmoid	16.37	77.26
LE ₃	FC (1×)	–	DwC (3 × 3)	ReLU	FC	AvgPool	Conv & Transpose	Sigmoid	8.93	77.19
Θ _{cap}	–	–	DwC (3 × 3)	ReLU	FC	AvgPool	Conv & Transpose	Sigmoid	8.18	77.17
LE ₄	FC (2×)	–	–	ReLU	FC	AvgPool	Conv & Transpose	Sigmoid	5.56	76.64
LE ₅	FC (1.5×)	–	–	ReLU	FC	AvgPool	Conv & Transpose	Sigmoid	4.37	76.38
Θ _{lite}	FC (1×)	–	–	ReLU	FC	AvgPool	Conv & Transpose	Sigmoid	3.18	76.70
LE ₆	FC (1×)	–	–	–	–	AvgPool	Conv & Transpose	Sigmoid	1.31	75.56
ECA	–	–	–	–	–	AvgPool	Conv & Transpose	Sigmoid	0.56	75.03
Θ _{eff}	–	–	–	–	–	–	–	–	0	75.76

- **MPC-efficient:** Θ_{cap} , Θ_{lite} are 2.0× and 5.0× more efficient than the standard MHA, respectively. Θ_{eff} is the identity function and thus communication-free.
- **Layer-dependent:** To address the issue of irregular similarities across different MHA blocks, we suggest determining the attention adapter according to the capability-efficiency trade-off for each layer independently.

- **Load-balancing:** Θ_{cap} , Θ_{lite} , Θ_{eff} trade off the capability and efficiency differently. Concretely, Θ_{cap} is more capable of encoding cross-layer features and Θ_{eff} is more efficient, while Θ_{lite} is a balanced choice in-between.

By assigning the suitable adapter to each MHA block in C according to the capability-efficiency thresholds τ , τ_c , τ_t , we can subsequently encode and reuse the attention output $Z_{\ell-1}$ for layers $\ell \sim (\ell + |C| - 1)$ instead of computing them repeatedly, effectively reducing the inference cost of MHA in MPC.